

Lecture 09

Carbon Tariffs

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AEM 4510

Roadmap

1. What are carbon tariffs?
2. Basic trade theory: supply, demand, and tariffs
3. Adding pollution externalities to trade
4. Unilateral carbon taxes and leakage
5. Carbon tariffs to prevent leakage
6. Carbon tariffs alone
7. World carbon taxes
8. Different pollution intensities across countries

What are carbon tariffs?

Carbon tariffs

Carbon tariffs are tariffs imposed on imports based on their **carbon content**

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Why would we want carbon tariffs?

Understanding border carbon adjustments



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Carbon tariffs aim to "level the playing field" and reduce leakage

Carbon tariffs in practice

EU Carbon Border Adjustment Mechanism (CBAM):

- Phasing in 2023-2026, full implementation 2026
- Covers cement, iron/steel, aluminum, fertilizers, electricity, hydrogen
- Importers must purchase certificates matching EU ETS price

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UK CBAM: announced for 2027

US proposals: various bills proposed but none passed yet

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3. What complications arise in practice

The key insight: carbon tariffs are equivalent to a **consumption tax** plus a **production subsidy**

Basic trade theory

A simple trade model

Consider a small open economy (country i) that:

- Takes world prices as given
- Can import or export a homogeneous good

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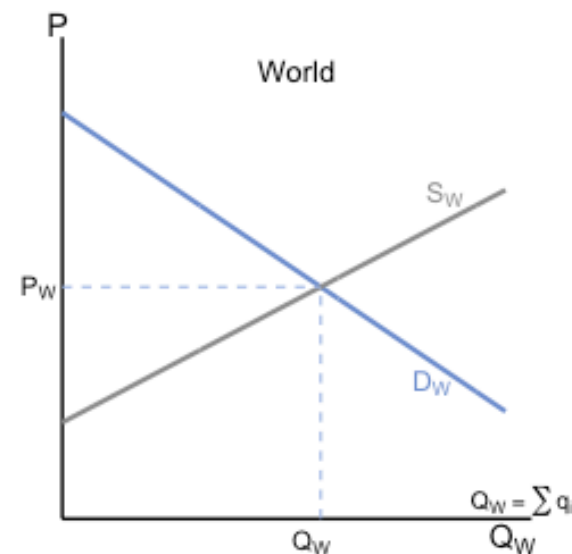
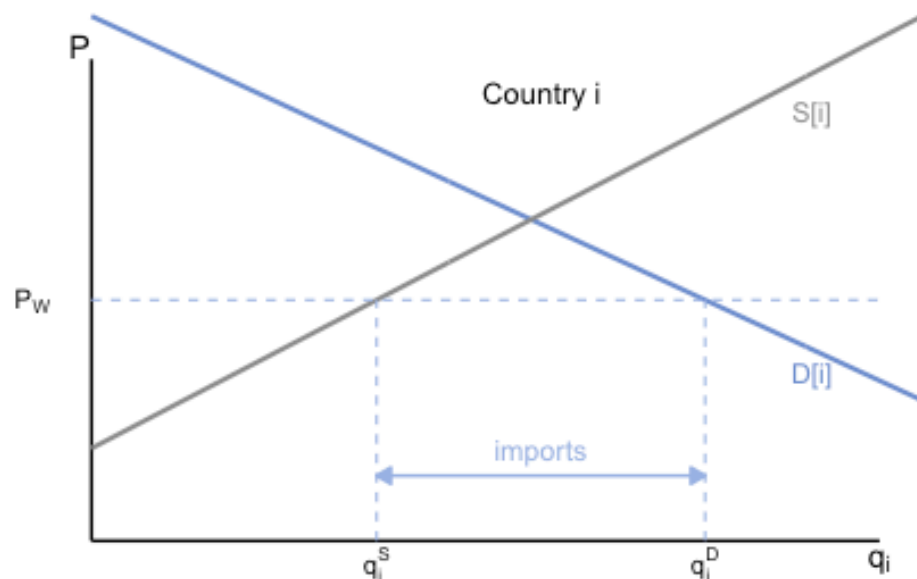
- Takes world prices as given
- Can import or export a homogeneous good

Key curves:

- S : domestic supply
- D : domestic demand
- P_W : world price
 - The social marginal benefit of the good

Domestic suppliers can't charge more than P_W , and won't charge less

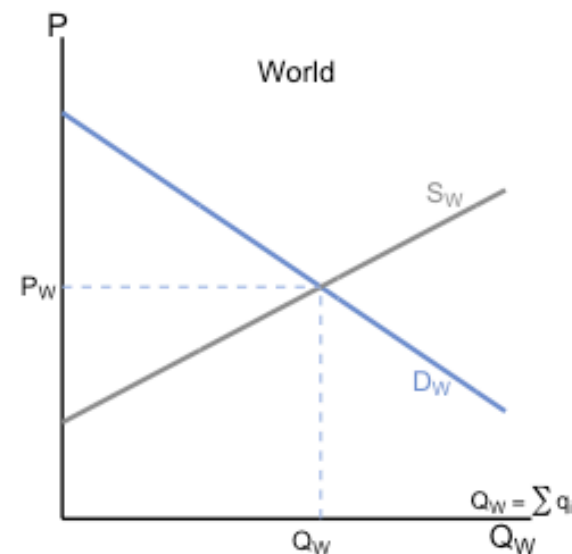
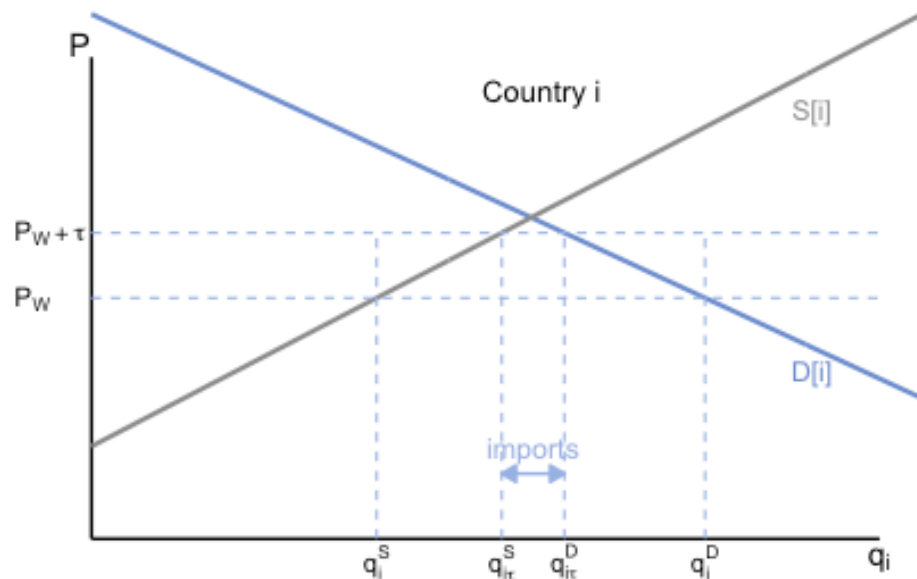
Free trade equilibrium



With free trade at price P_W :

- Domestic production: q_i^S
- Domestic consumption and imports: q_i^D and $q_i^D - q_i^S$

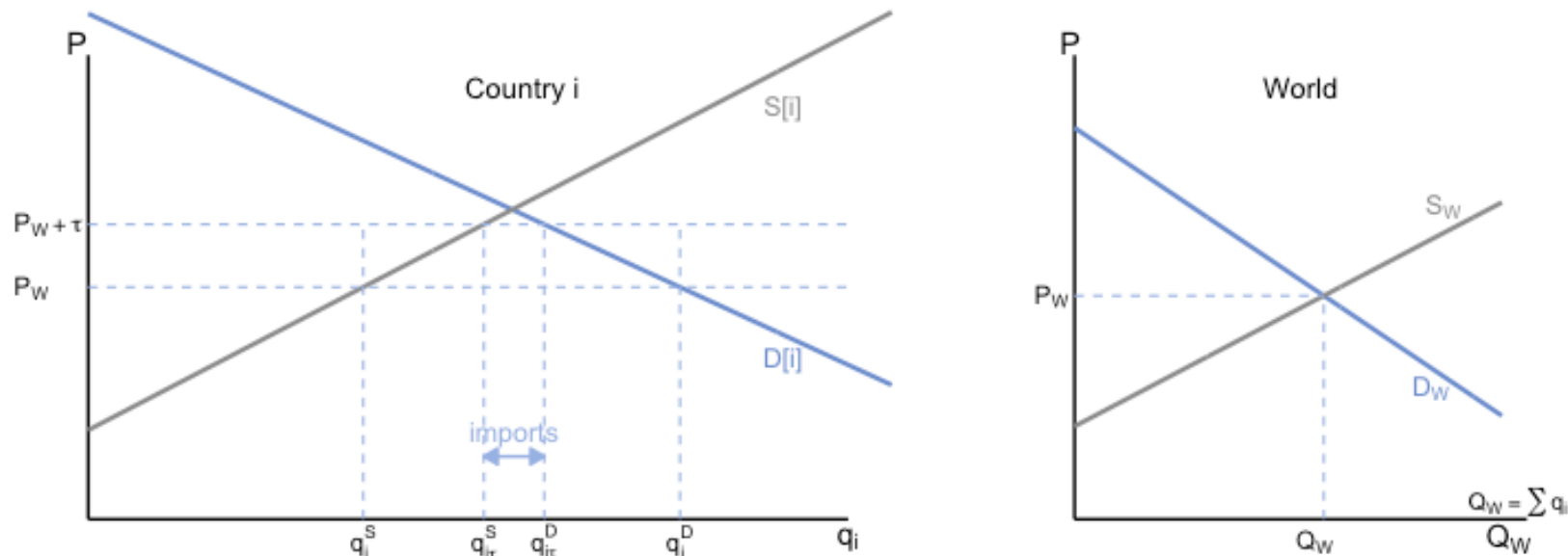
Adding a tariff



Suppose we implement an import tariff τ :

- Price of imports increases to $P_W + \tau$
- Domestic suppliers can now charge $P_W + \tau$

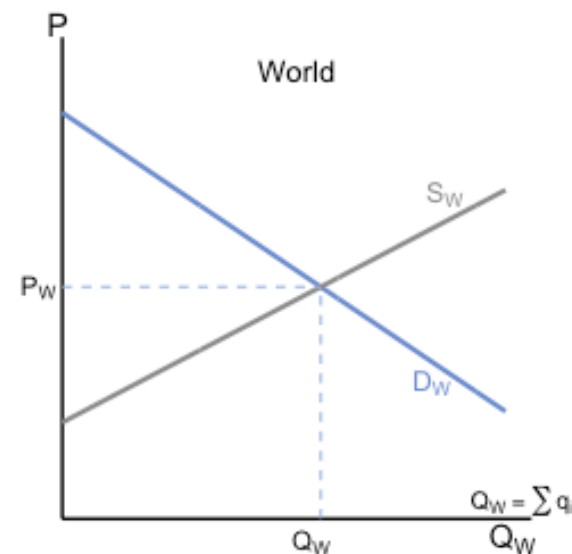
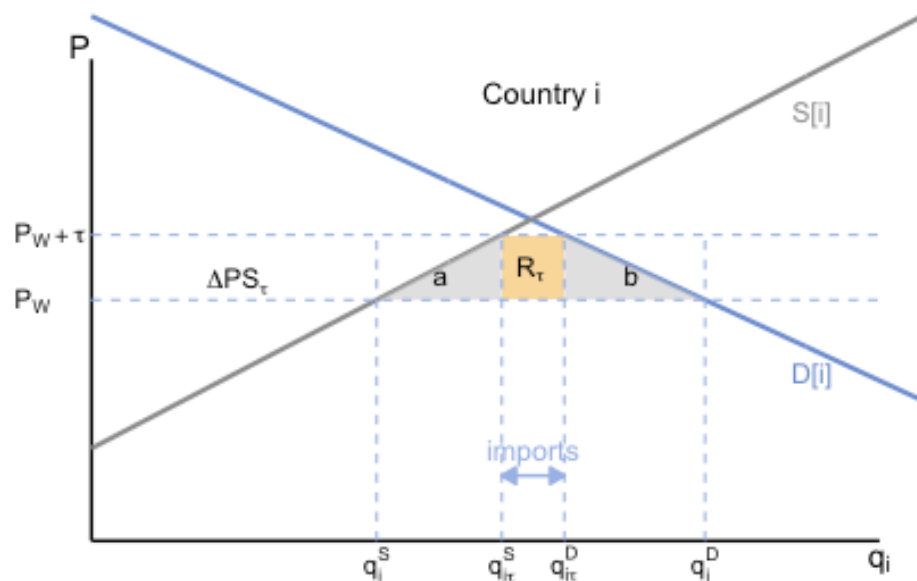
Adding a tariff



A tariff τ raises the equilibrium domestic price to $P_W + \tau$:

- Production rises to $q_{i\tau}^S$ and consumption falls to $q_{i\tau}^D$
- Imports fall to $q_{i\tau}^D - q_{i\tau}^S$

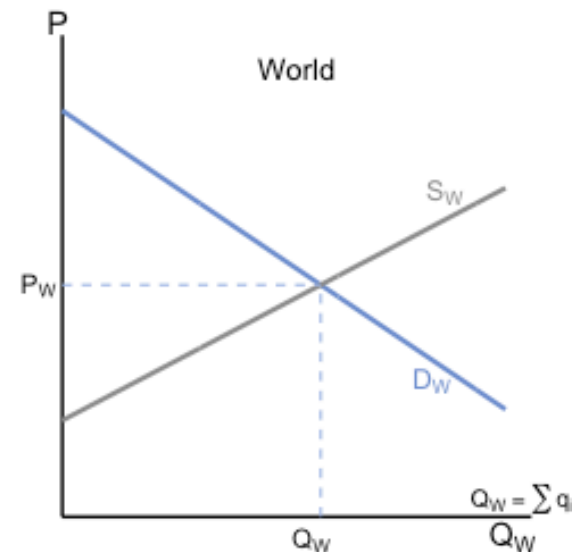
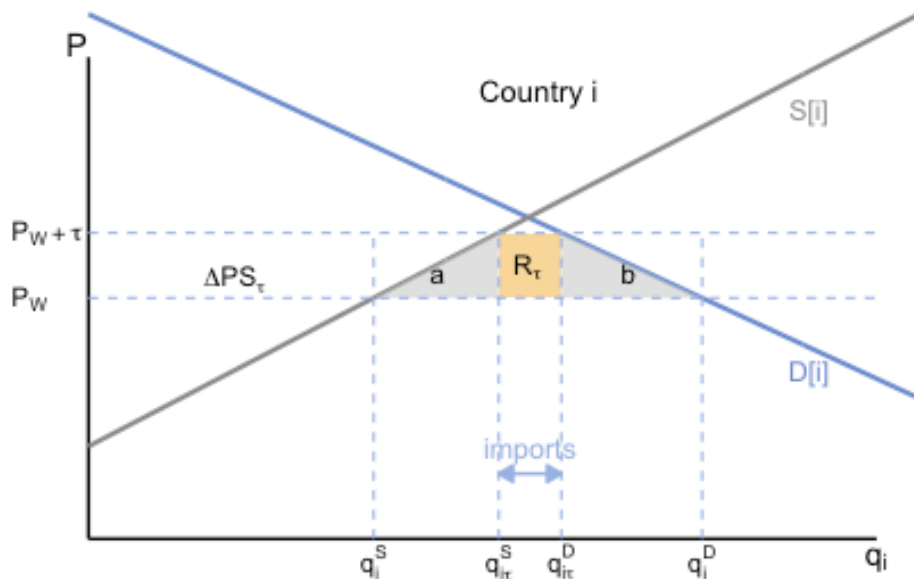
Tariff revenue



The tariff creates:

- Revenue: $R(\tau) = \tau(q_{i\tau}^D - q_{i\tau}^S)$
- Deadweight loss: triangles $a + b$

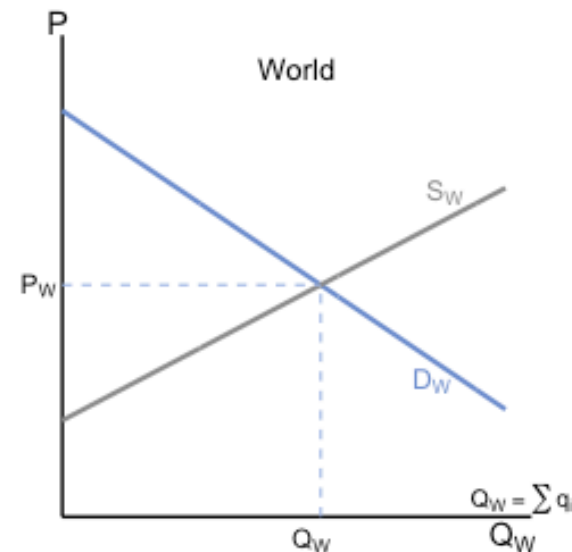
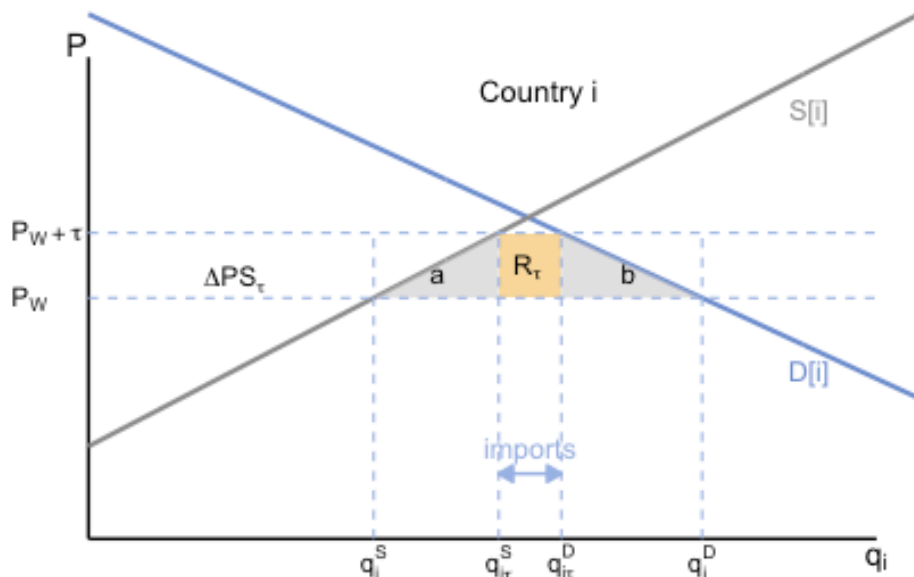
Deadweight loss



Triangle a is a production distortion:

- The tariff shifts production from lower-cost foreign supply to higher-cost domestic production: MC ($S[i]$) is higher than the MB (P_W)

Deadweight loss



Triangle b is a consumption distortion:

- The tariff eliminates units consumers value above the world price but below the tariff-inclusive price: MB ($D[i]$) is higher than the MC (P_W)

Adding pollution

Production creates pollution

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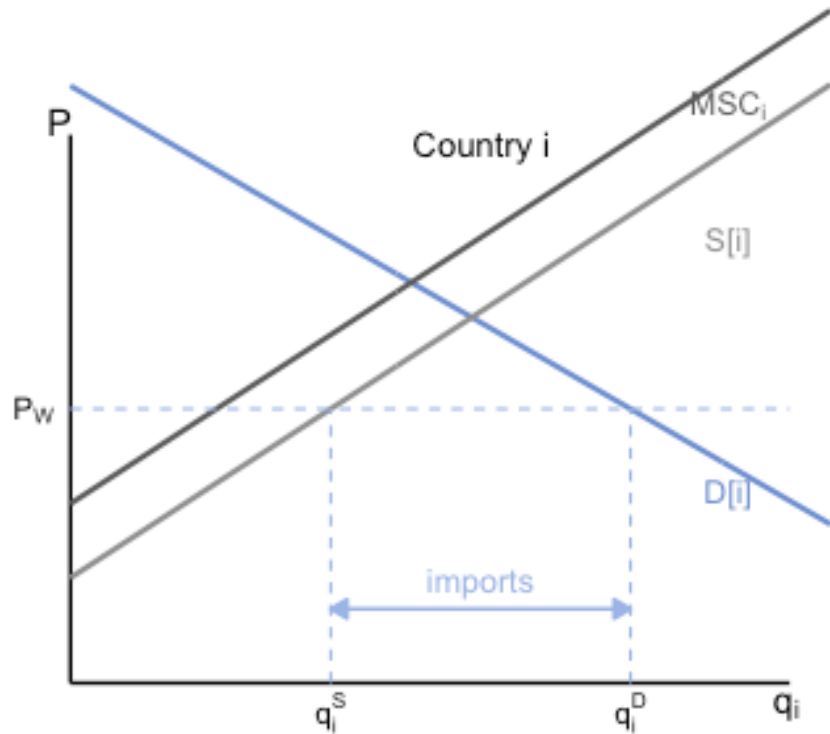
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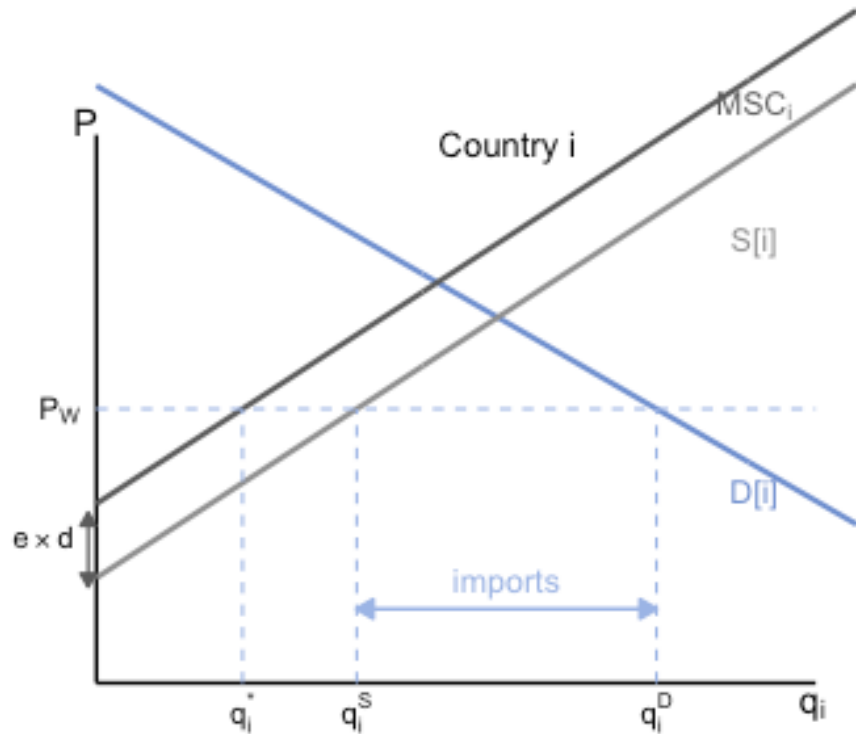
MSC lies above the private supply curve by the external cost $e \times d$

Free trade with pollution



Free trade output is pinned down by domestic demand and supply at the world price

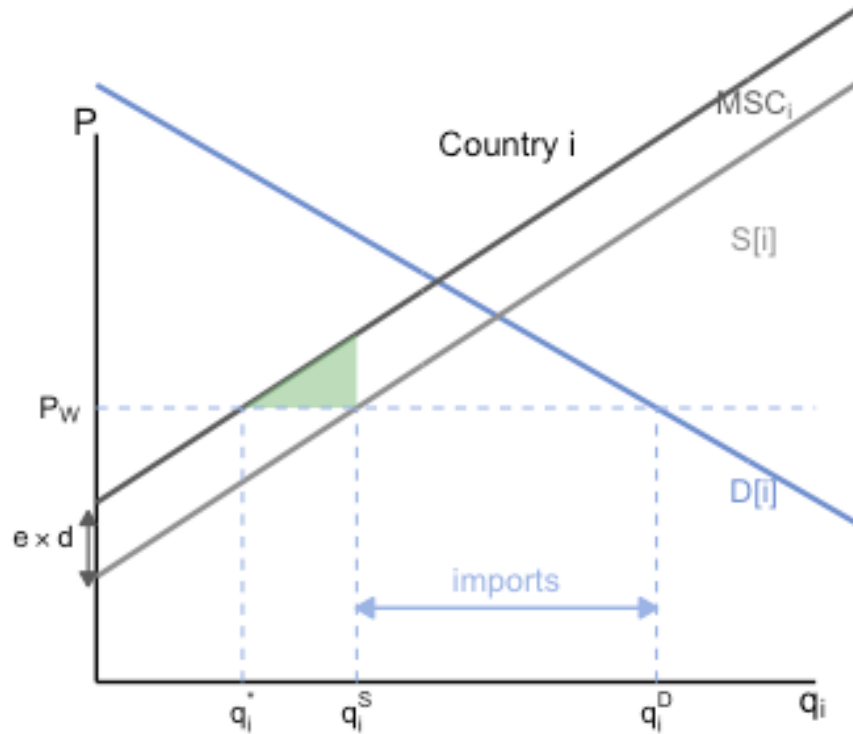
Free trade with pollution



Country i 's MSC curve lies above its private supply curve

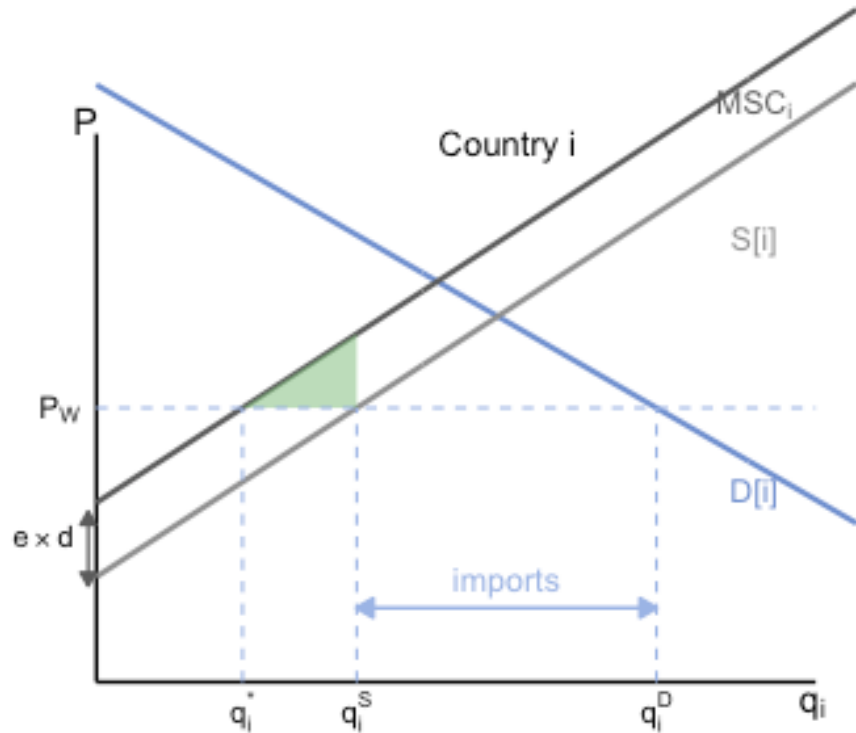
The vertical gap is the external cost: $e \times d$

Free trade with pollution



The green region is the domestic deadweight loss from producing beyond the efficient output level

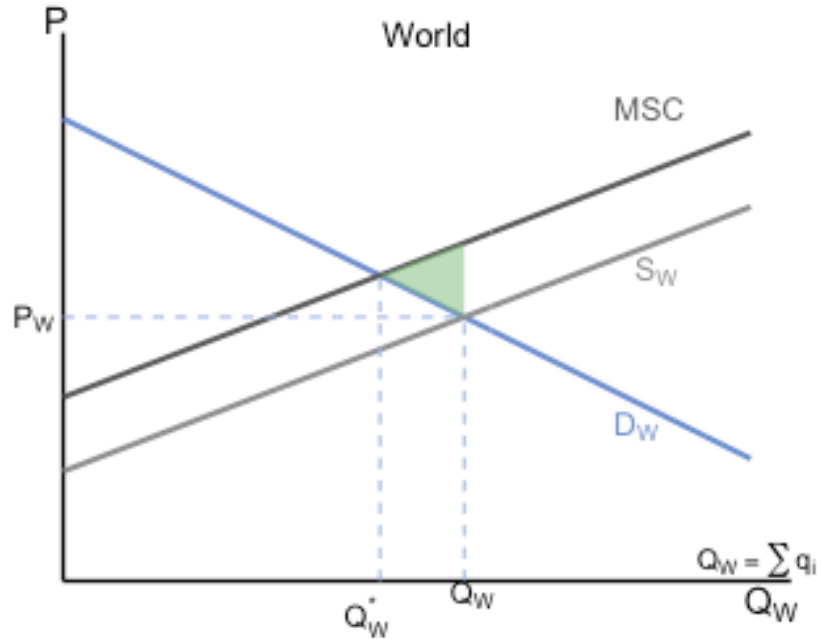
Free trade with pollution



At the world price P_W , marginal benefit is pinned down by world demand, while marginal social cost is given by MSC

That triangle is the set of units for which marginal benefit exceeds private supply cost but is below marginal social cost

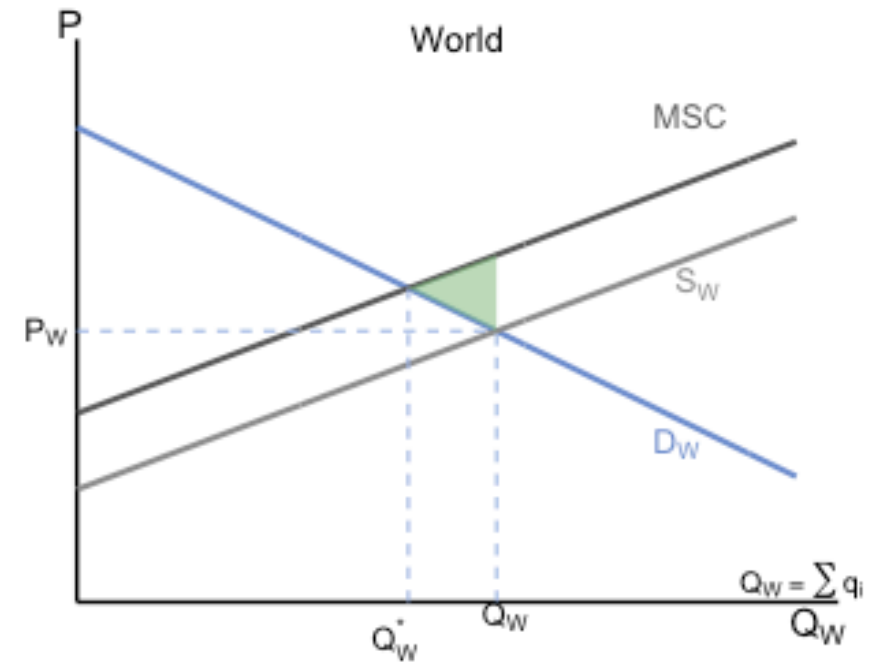
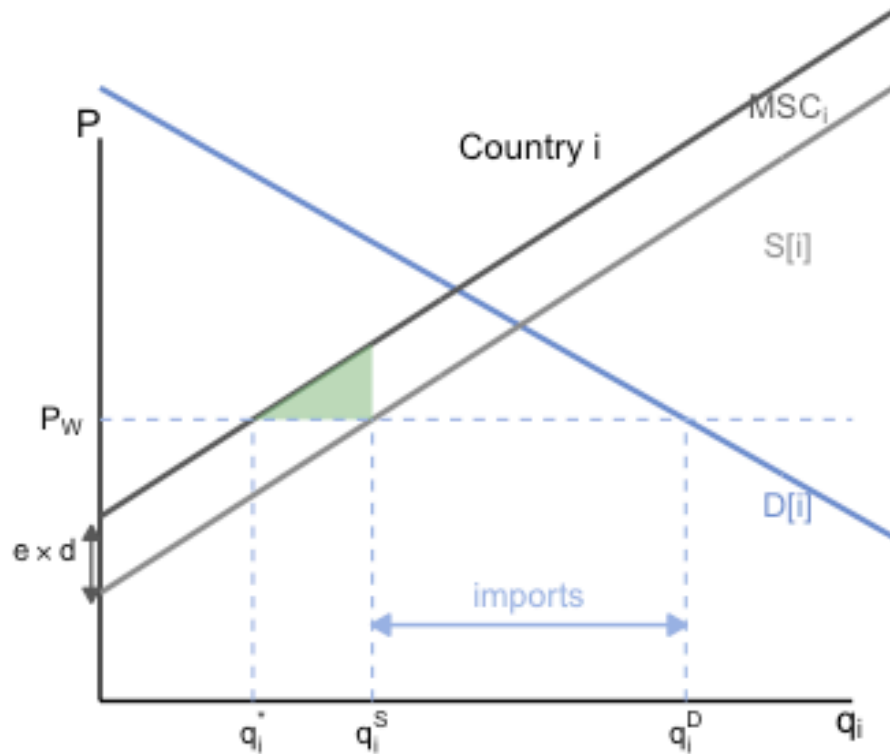
Global perspective



From a global perspective:

- Free-trade output is where $D = S$
- Efficient output is where $D = MSC$
- The green region is the global welfare gain from cutting output from Q_W to Q_W^*

Country and global perspectives



Global DWL on the right is the sum of each country's DWL on the left

The externality problem

With a global pollutant like CO_2 :

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But what if only **one country** acts?

Unilateral carbon tax

Country i imposes a carbon tax

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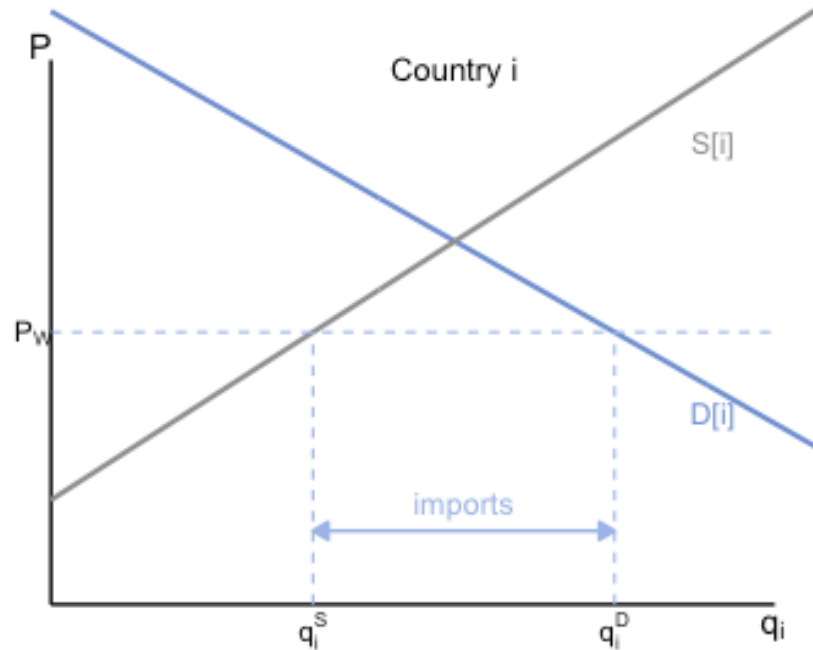
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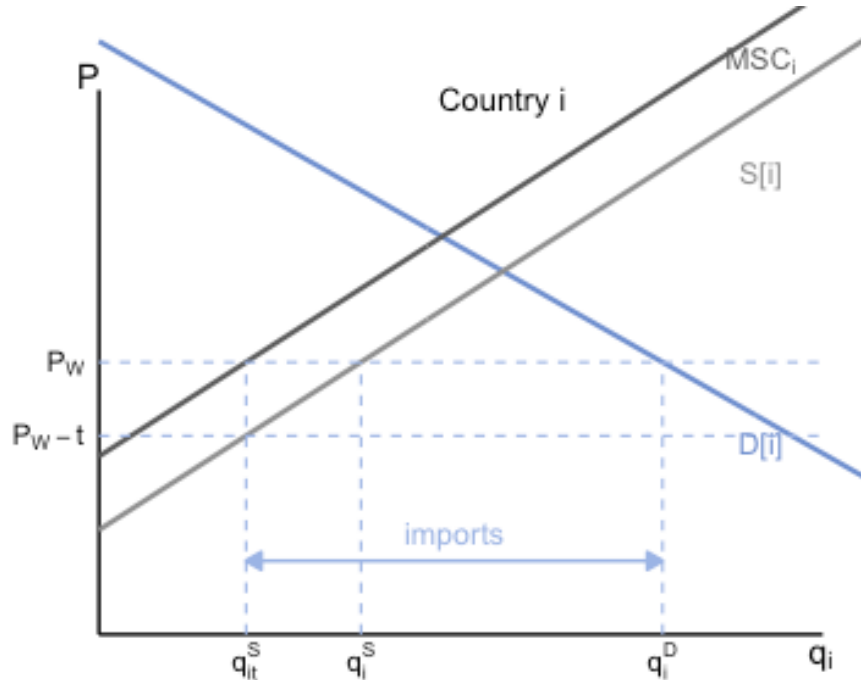
But the **world price doesn't change** (small open economy)

Effects of unilateral carbon tax



Before the tax, the domestic market is at the free-trade benchmark with price P_W

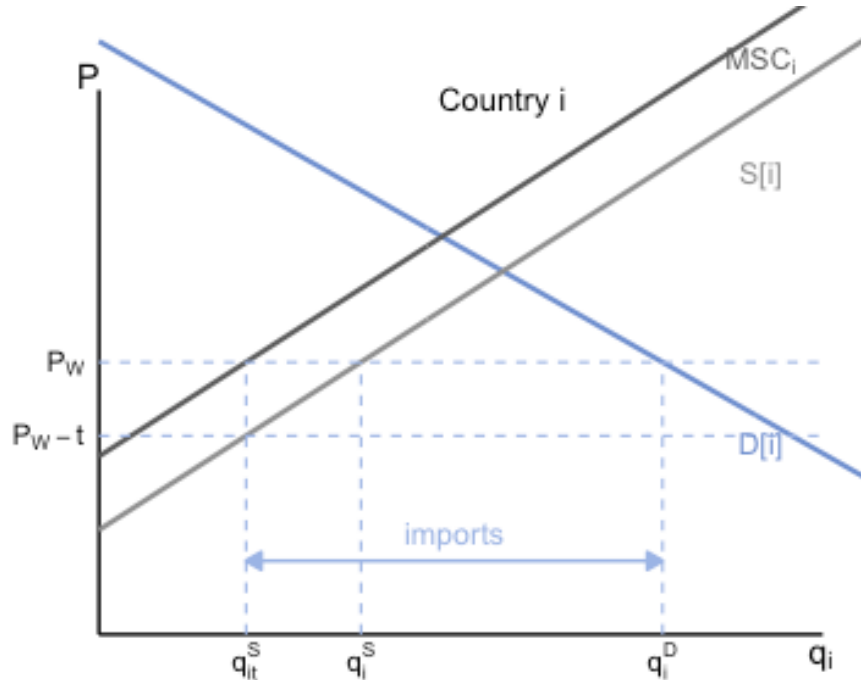
Effects of unilateral carbon tax



After the tax:

- Producers receive the net-of-tax price $P_W - t$
- Because t equals the external damage, domestic output falls to q_{it}^S
- Consumers still face the world price P_W

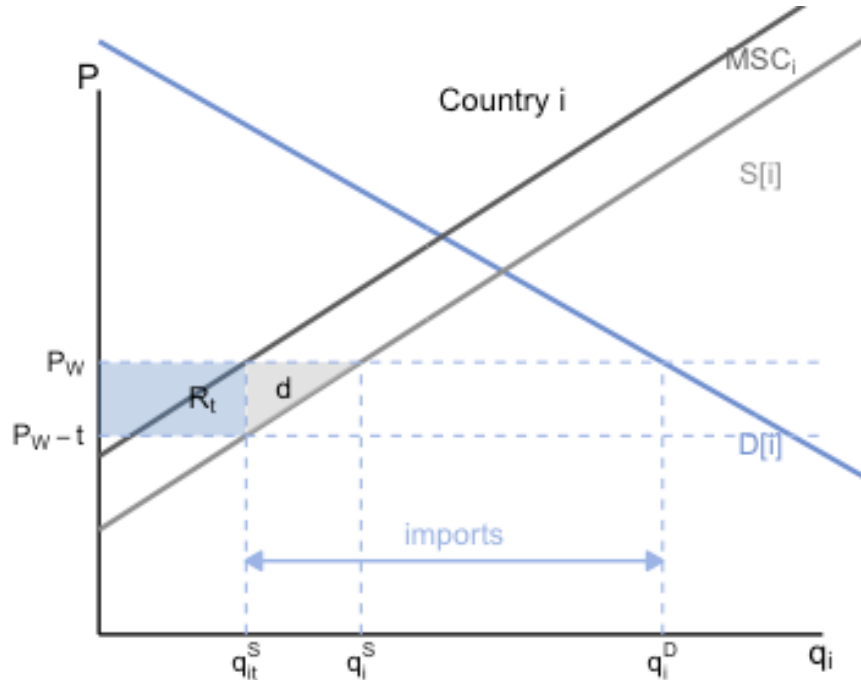
Effects of unilateral carbon tax



Leakage comes next:

- Consumers in country i still demand q_i^D at price P_W
- Imports rise enough to offset the fall in domestic output
- Foreign production replaces home production: that is leakage

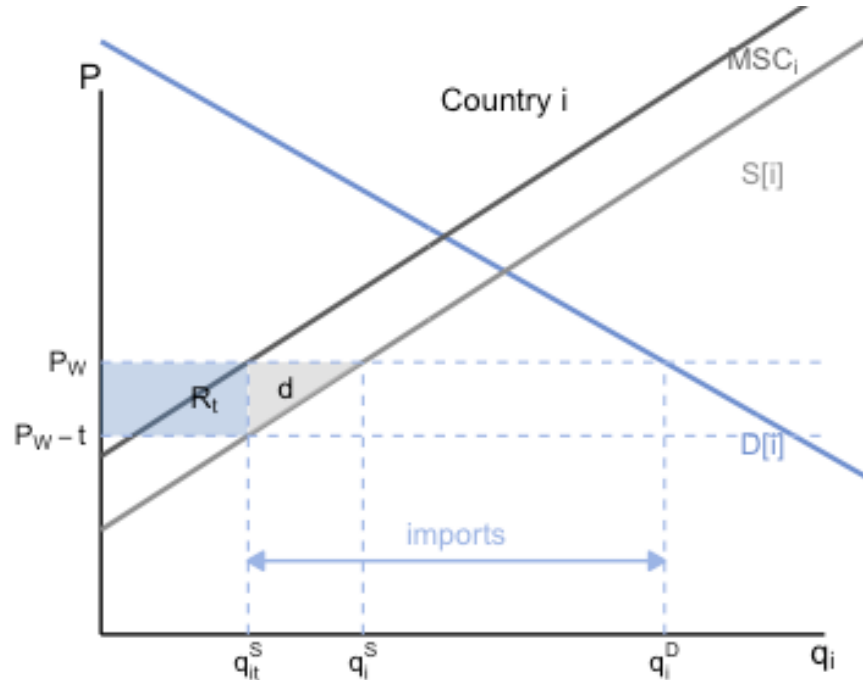
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From country i 's perspective:

- Consumers are no better or worse off: they still pay P_W and consume q_i^D
- The government gains revenue R_t

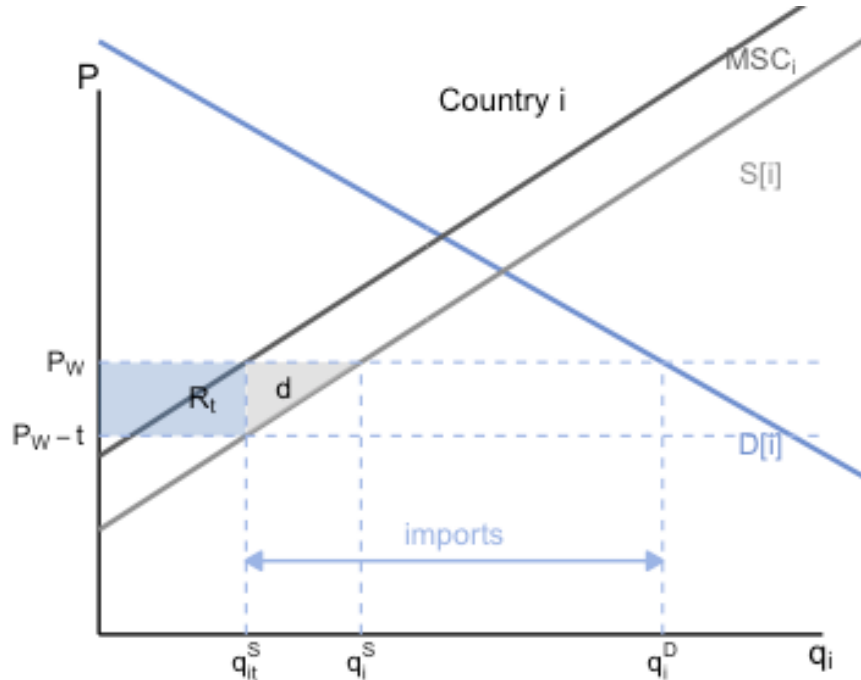
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From country i 's perspective:

- Producers lose producer surplus equal to $R_t + d$

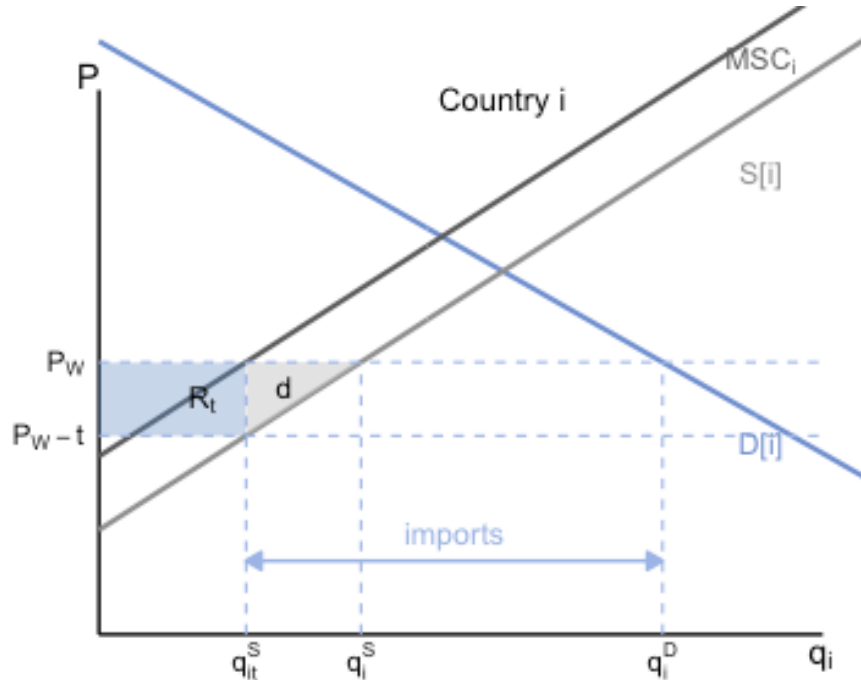
Effects of unilateral carbon tax



Why do producers lose $R_t + d$?

- They receive a lower net-of-tax price on the units they still sell
- They also stop selling the units between q_{it}^S and q_i^S

Effects of unilateral carbon tax

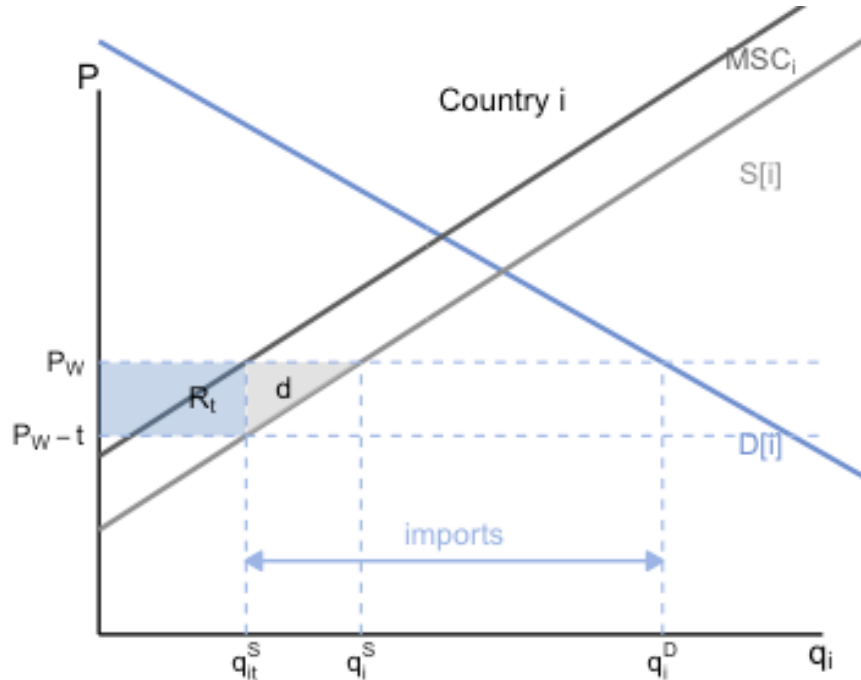


Why is d the country's deadweight loss?

- R_t is only a transfer from producers to the government, so it stays inside country i
- Net national effect:

$$+R_t - (R_t + d) = -d$$

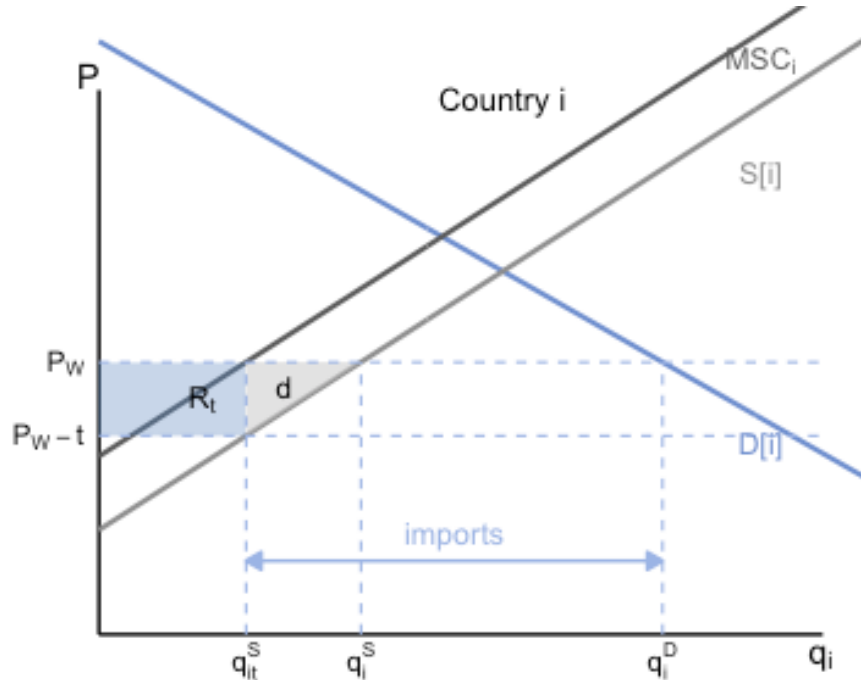
Effects of unilateral carbon tax



Why does leakage make d the relevant national loss?

- The triangle d is the pure private loss left after netting out the transfer R_t
 - Goods not produced where $PMC (S[i]) < SMB (P_W)$
- Consumers in country i are unchanged and imports replace the lost domestic output

Effects of unilateral carbon tax



Why does leakage make d the relevant national loss?

- The tax reduces home production, but foreign production rises 1-for-1: that is leakage
- The environmental gain is offset abroad, so country i is left with the private DWL d

Unilateral carbon tax effects

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- World price is unchanged at P_W so quantity demanded domestically for the good does not change
- Imports from abroad fill in the difference
- Production and emissions increase abroad

Leakage undermines the tax

With identical carbon intensities and a small country implementing the tax (so P_W is fixed):

- 1 unit fewer domestic production = 1 unit more foreign production
- **Net global emissions unchanged!**

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The carbon tax:

- Generates tax revenue R_t (rectangle)
- Creates deadweight loss d (triangle) from production distortion
- Achieves **zero environmental benefit**

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Country i bears costs but gets no environmental gain

Leakage is a rate, not a constant

Economists often summarize leakage with the **leakage rate**:

$$\text{Leakage rate} = \frac{\Delta E_j}{-\Delta E_i}$$

where $\Delta E_i < 0$ is the change in home emissions and $\Delta E_j > 0$ is the change in foreign emissions

Leakage is a rate, not a constant

$$\text{Leakage rate} = \frac{\Delta E_j}{-\Delta E_i}$$

Benchmarks:

- 0: no foreign offset
- Between 0 and 1: global emissions still fall, but less than the domestic reduction
- 1: one-for-one offset, as in this small-open-economy example with identical intensities
- Greater than 1: possible if output shifts to dirtier foreign producers

What determines leakage?

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Leakage is **larger** when the world price moves less in response to the tax:

- The taxing country is small in world markets, so its supply cut hardly changes the world price
- With little price movement, domestic demand doesn't change and imports replace the lost domestic output

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Leakage is **larger** when foreign replacement is elastic or dirty:

- Foreign supply is elastic and trade costs are low, so firms abroad can expand output quickly when home production falls
- Foreign production is more emissions-intensive than home production ($e_j > e_i$), so each unit shifted abroad offsets more of the home emissions reduction

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Leakage is **smaller** when policy reduces quantity demanded by increasing the world price:

- The taxing country is large in world markets, so its supply cut changes the world price
- With price increases, domestic quantity demanded falls
- Then fewer imports are needed to fill the gap, so there is less of an emissions increase abroad, and greater global emissions reductions

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What determines leakage?

Leakage is **smaller** when foreign expansion is constrained or cleaner:

- If foreign producers supply is inelastic, firms abroad are limited in how much they can increase production
- If home production is dirtier than foreign production ($e_i > e_j$), shifting output abroad offsets less of the domestic abatement and can even lower global emissions

Carbon tax + carbon tariff

Adding a carbon tariff

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One potential way to address this is a **carbon tariff** of rate τ : a tax on imports proportional to their carbon content

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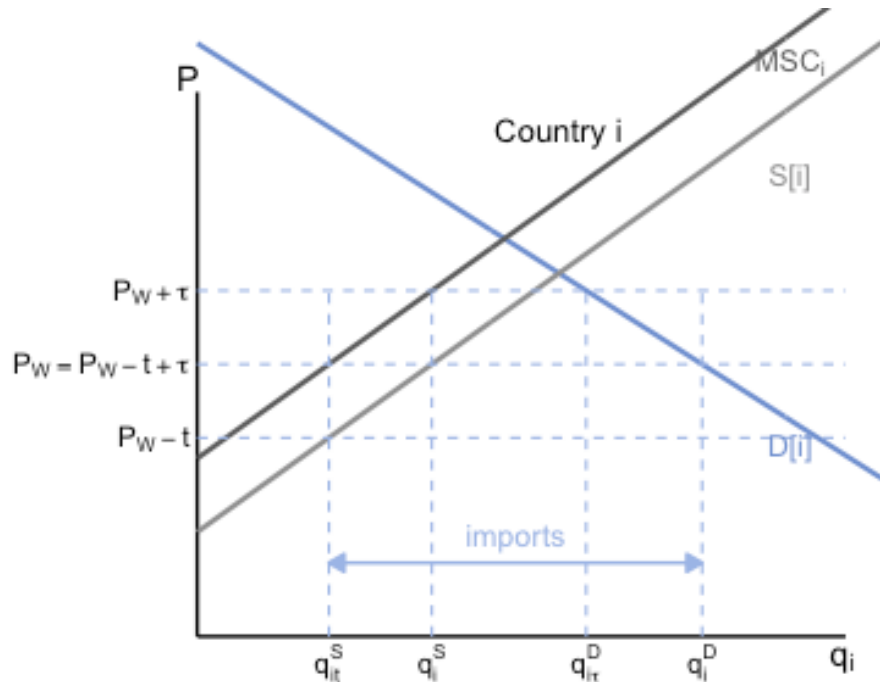
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Here we will tax imports at the same rate as domestic production:

$$\tau = t = e \times d$$

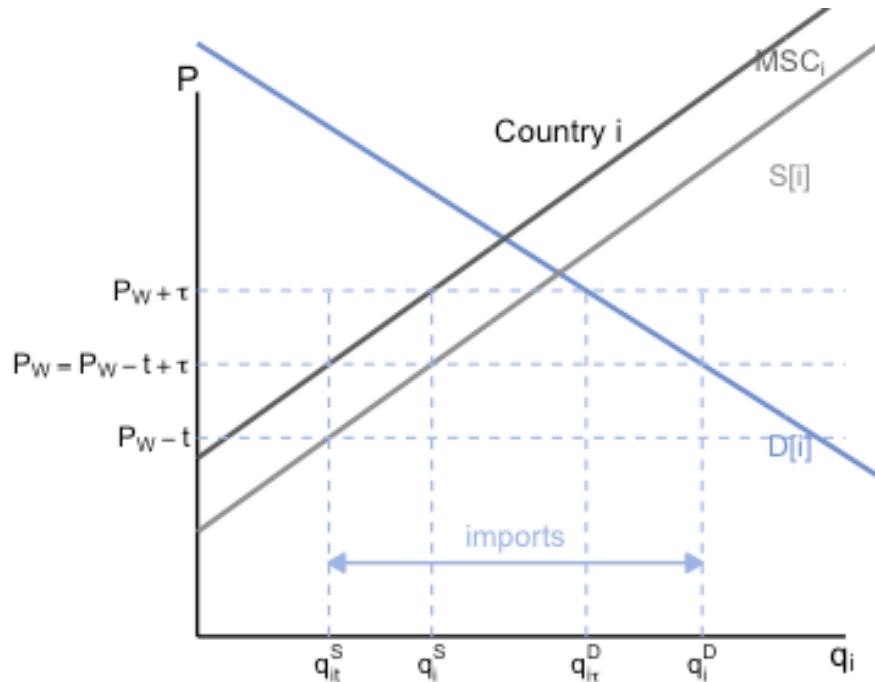
Adding a carbon tariff



Start from the carbon tax alone:

- Producers net $P_W - t$
- Home output falls to q_{it}^S
- Consumers still pay P_W and buy q_i^D

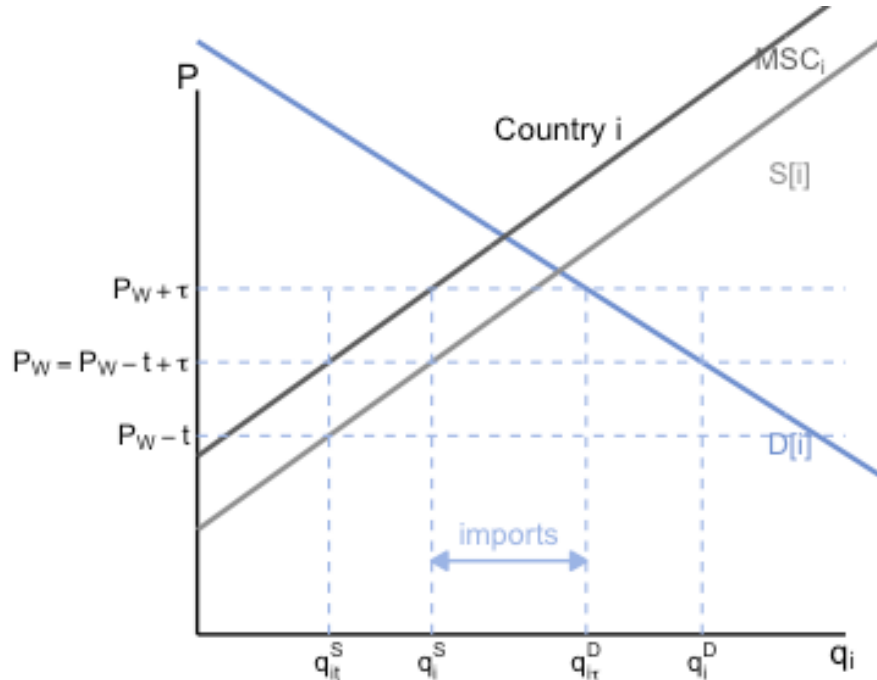
Adding a carbon tariff



Why is leakage the problem here?

- Imports replace the lost domestic output
- So home emissions fall, but foreign production rises
- The tax changes domestic supply, not consumption
- Since consumption doesn't change, there must be more imports

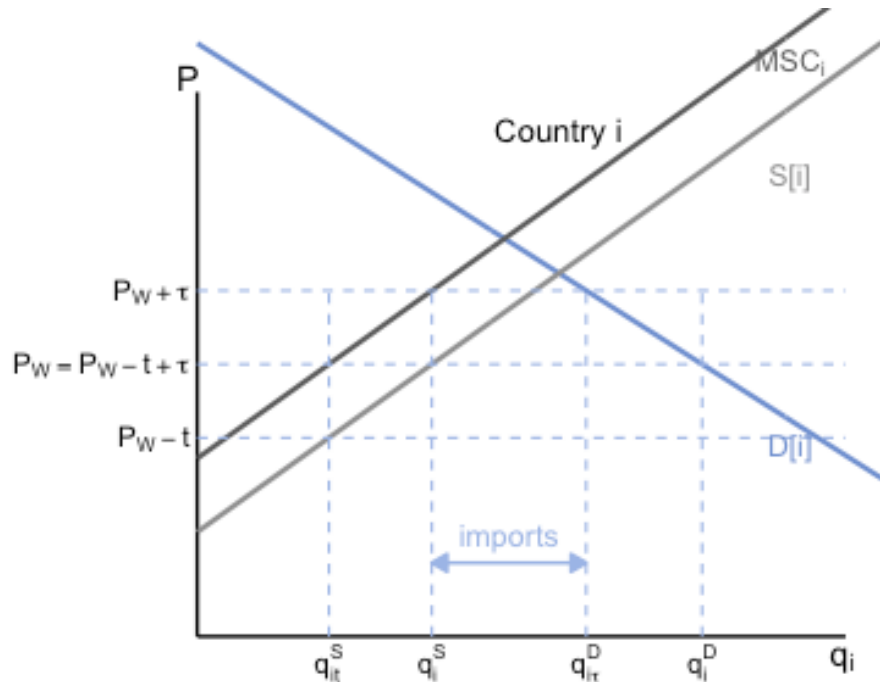
Adding a carbon tariff



Now add a carbon tariff $\tau = t$ on imports:

- Producers can now charge $P + \tau$ so consumers pay $P_W + \tau$
- Domestic producers only net $P_W + \tau - t = P_W$ after the carbon tax payment
- The tax and tariff offset for producers so domestic supply returns to q_i^S

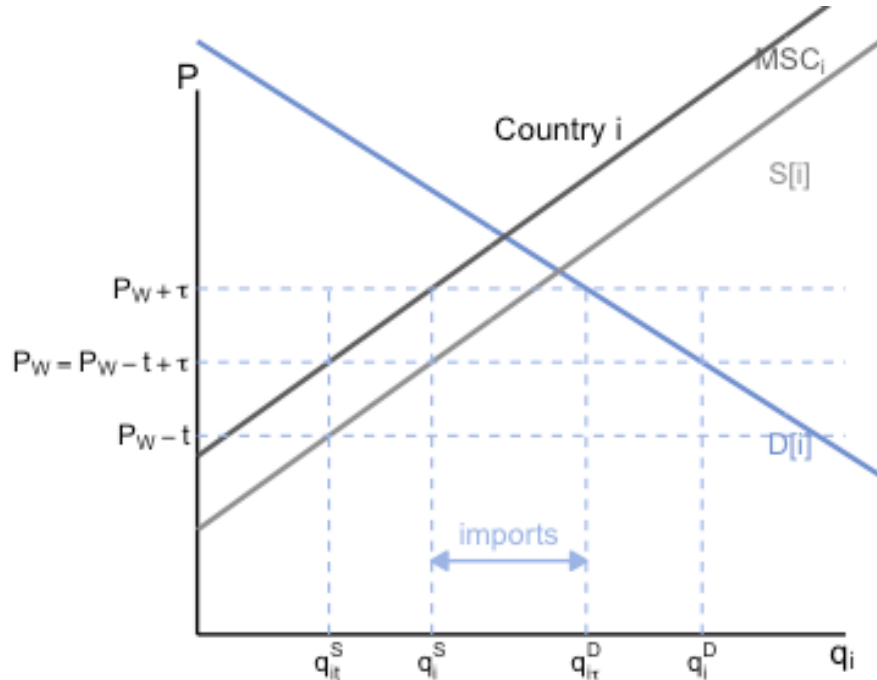
Adding a carbon tariff



What changes in demand and trade?

- Consumers pay $P_W + \tau$ so domestic quantity demanded falls to $q_{i\tau}^D$
- Imports shrink from $q_i^D - q_{it}^S$ to $q_{i\tau}^D - q_i^S$

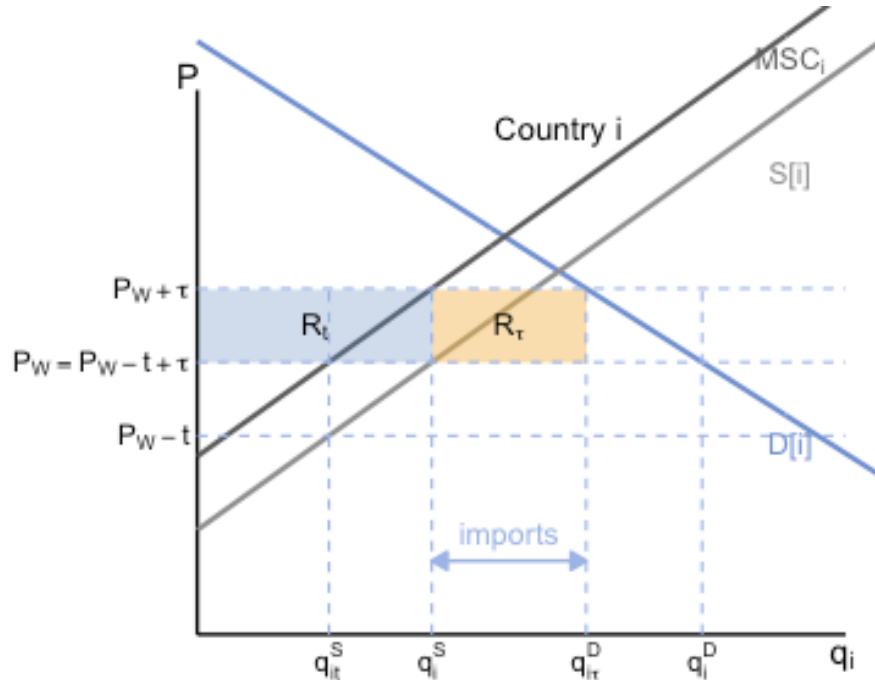
Adding a carbon tariff



Total supply and demand effects of tax+tariff:

- No net effect on producers since their quantity is the same as unregulated
- Consumers face a higher price $P_W + \tau$, reduce consumption, and reduce imports

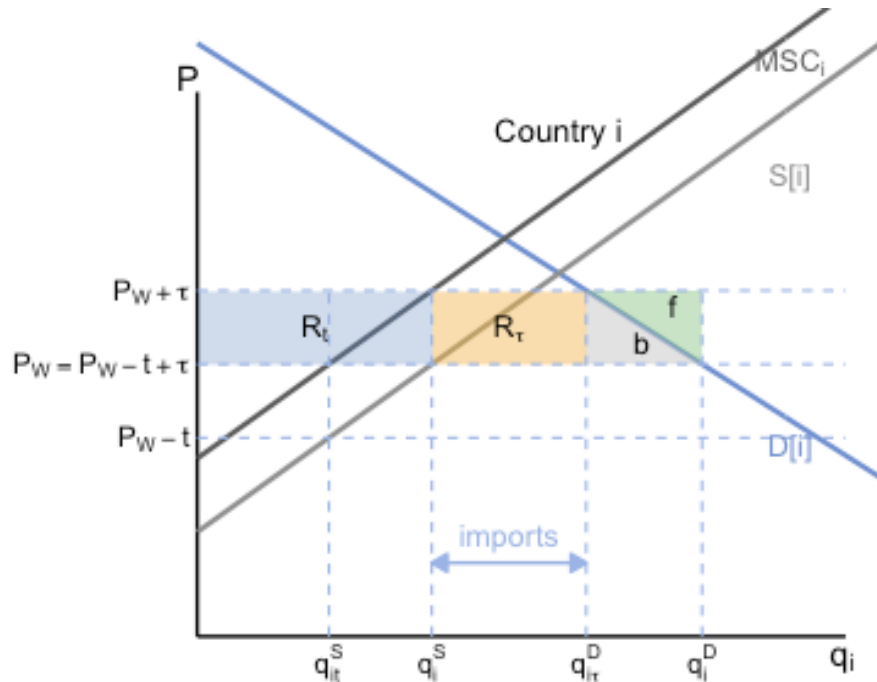
Adding a carbon tariff



Revenue effects:

- The domestic carbon tax still raises R_t
- The tariff raises R_τ on remaining imports
- Both are transfers inside country i : tax R_t paid by producers, tariff R_τ paid by consumers

Adding a carbon tariff



Welfare and environment:

- b is the consumption deadweight loss from the higher price
- $b + f$ is the environmental gain from lower world output and emissions
- Tax + tariff improves social welfare by f

Recap

With only a tax:

- Domestic production/emissions become more expensive
- Consumers can easily switch to imports: "leakage"
- We taxed production but received no environmental benefits

Adding a carbon tariff:

- We are now making foreign production/emissions more expensive for our domestic consumers
- Consumers can no longer as easily switch to imports
- Emissions fall because they are correctly priced (good!)
- Consumption falls (bad!)

The key insight

With identical carbon intensities everywhere:

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Carbon tax + carbon tariff = Consumption tax

The key insight

With identical carbon intensities everywhere:

Carbon tax + carbon tariff = Consumption tax

Why?

- Tax on domestic production raises domestic costs
- Tariff on imports raises import costs
- Both domestic and imported goods face the same carbon price
- Equivalent to taxing consumption of the carbon-intensive good

Carbon tariff alone

What if we only use a tariff?

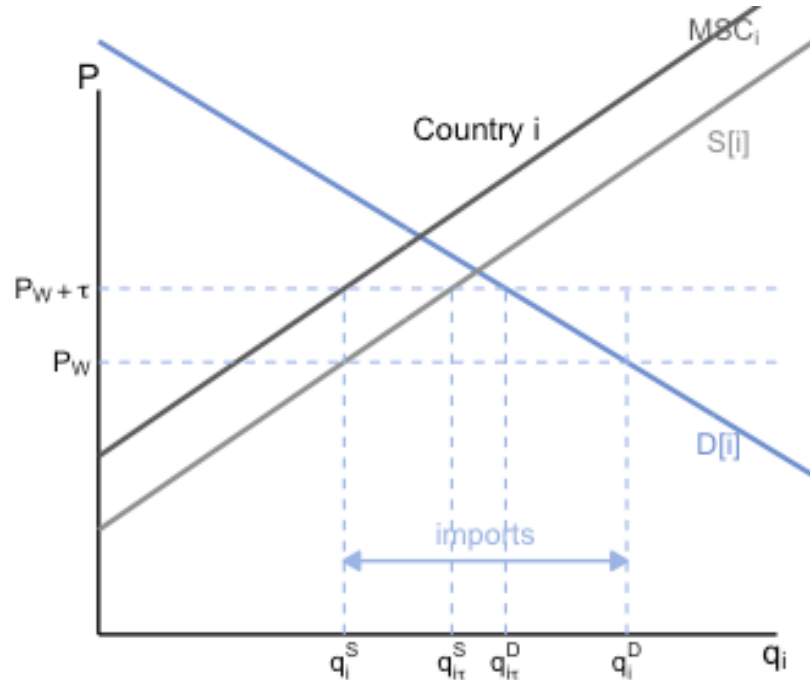
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What does this accomplish?

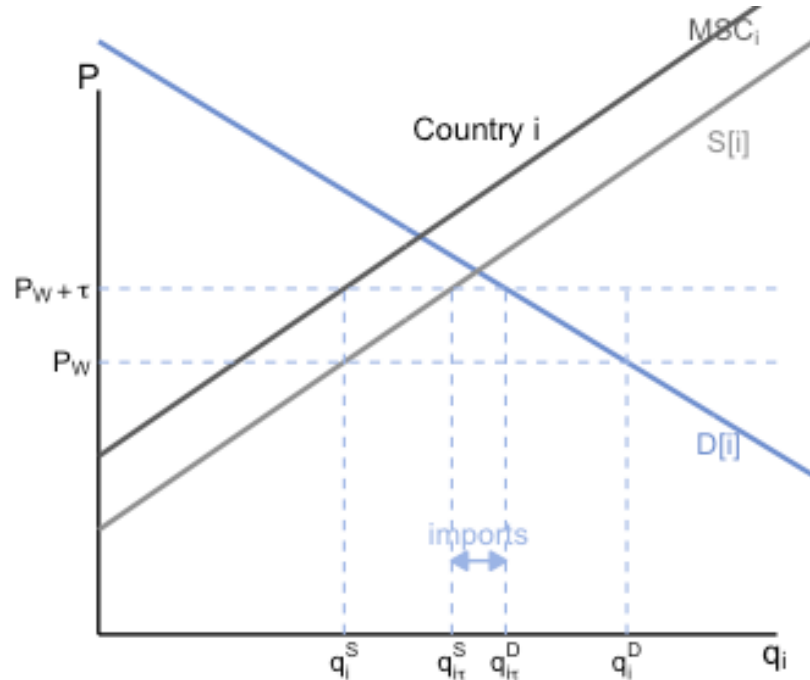
Carbon tariff alone



Start from free trade:

- Consumers and producers in country i both face P_W
- Domestic output is q_i^S and consumption is q_i^D
- Imports fill the gap $q_i^D - q_i^S$

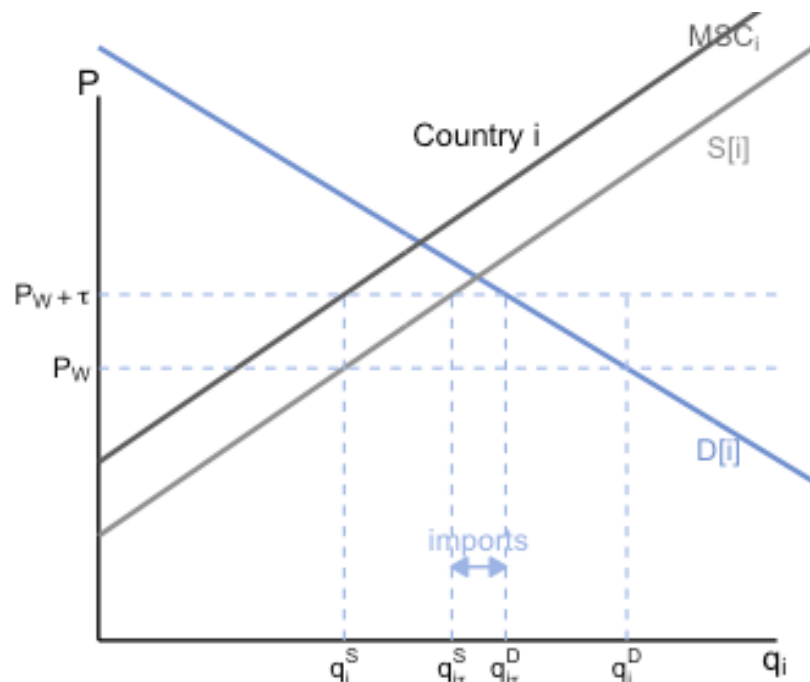
Carbon tariff alone



Now impose a carbon tariff τ with no domestic carbon tax:

- Consumers pay $P_W + \tau$
- Domestic producers also receive $P_W + \tau$
- So consumption falls and home output rises

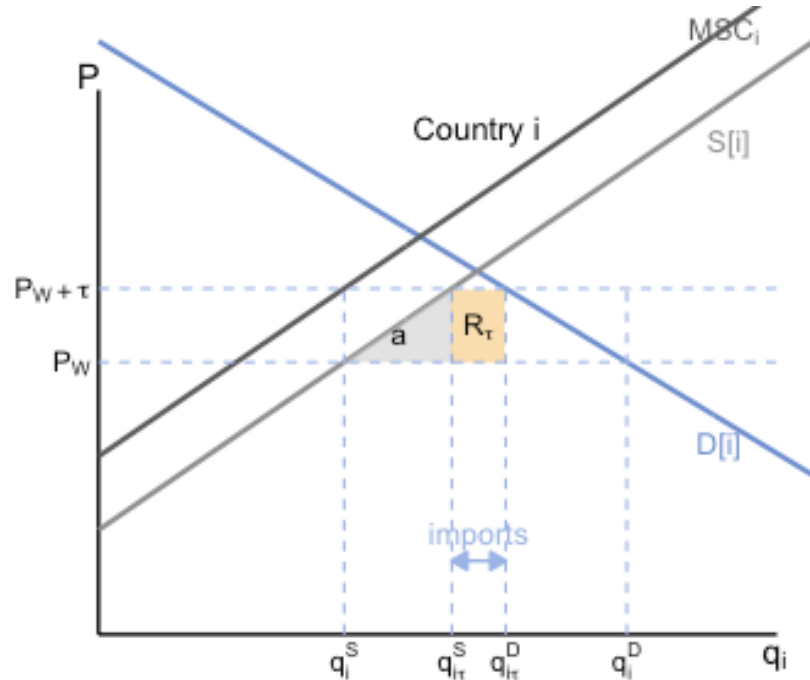
Carbon tariff alone



Why is this different from tax + tariff?

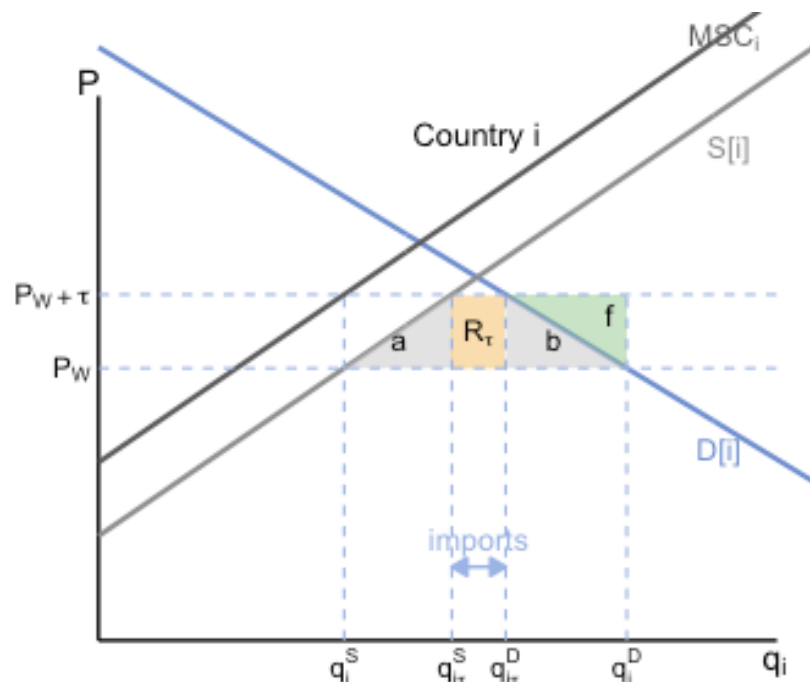
- The tariff alone taxes consumption but also protects domestic producers
- Output rises to $q_{i\tau}^S$ and demand falls to $q_{i\tau}^D$
- Imports shrink because both margins move

Carbon tariff alone



- The government collects tariff revenue R_τ on remaining imports
- The tariff pushes some production away from cheaper foreign suppliers

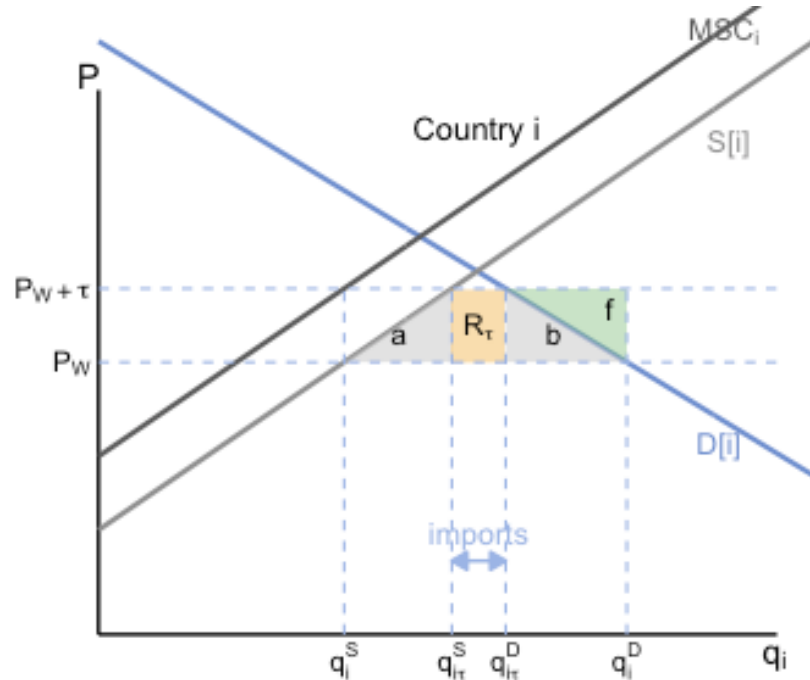
Carbon tariff alone



Welfare and environment:

- a is the production deadweight loss from replacing cheaper imports with costlier home output
- Reduced consumption creates environmental gains of $b + f$, offsetting the consumption loss b

Carbon tariff alone

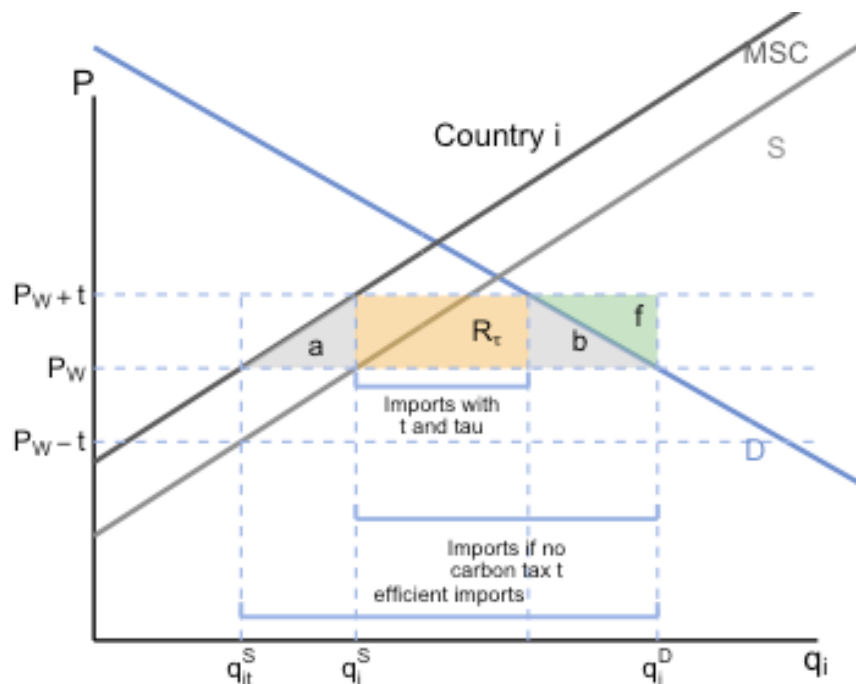


Welfare and environment:

- Net social effect is $(b + f) - (b + a) = f - a$, so a tariff alone mixes climate gains with a subsidy to domestic pollution

World carbon tax

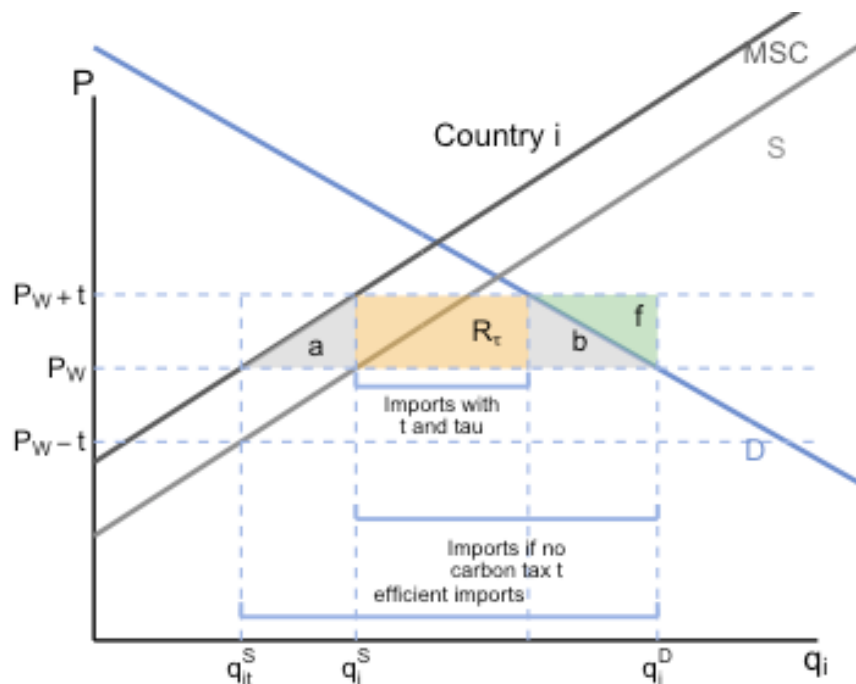
World carbon tax



Suppose the rest of the world already prices carbon so P_W internalizes the externality *abroad*:

- P_W is then where abroad $MSC =$ abroad D
- i produces at q_i^S
- At q_i^S : $MSC >$ world MB given by P_W leading to DWL a

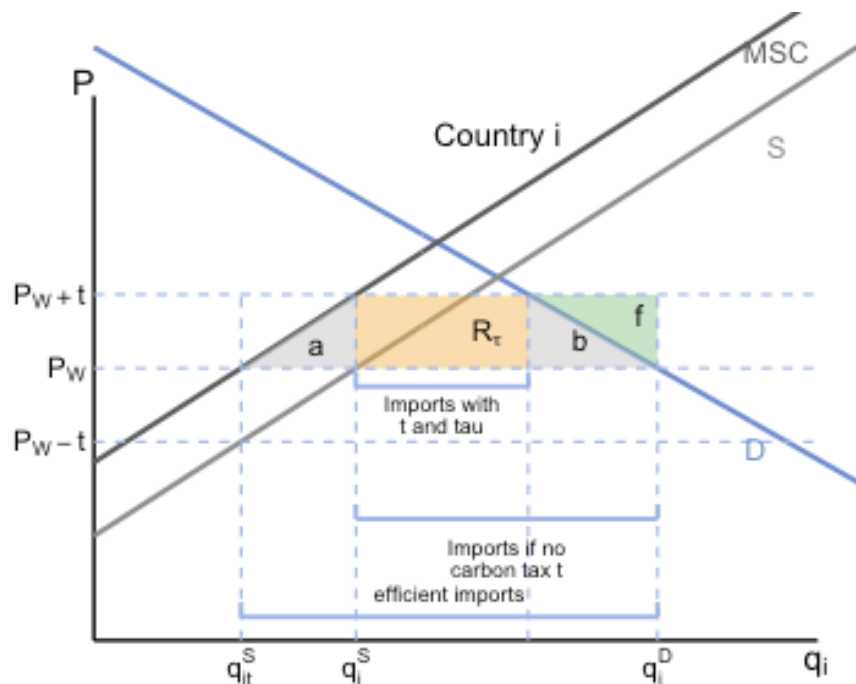
World carbon tax



If country i also imposes the carbon tax t :

- Producers receive net-of-tax price $P_W - t$
- Supply falls from q_i^S to q_{it}^S
- Imports rise to $q_i^D - q_{it}^S$ to make up the shortfall given quantity demanded does not change given world price P_W

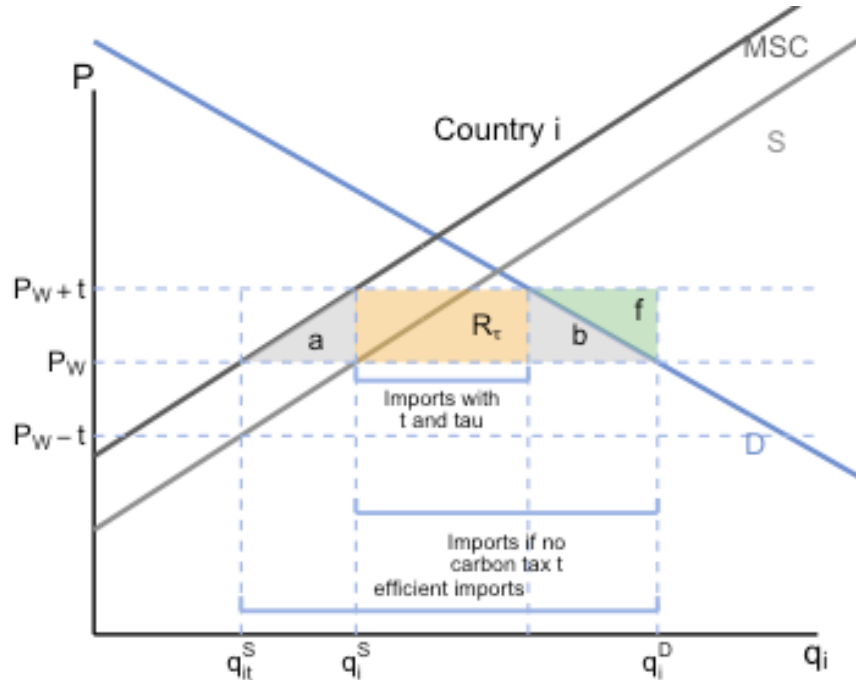
World carbon tax



Why are those extra imports not leakage?

- Foreign producers already face the same carbon price
- That foreign carbon price led to leakage *into* country i
- Those imports reverse leakage from the rest of the world to country i
- The DWL a disappears

World carbon tax



At that point the allocation is efficient:

- Every country has internalized the externality
- Global marginal social cost equals the marginal benefit of the last unit consumed
- A carbon tariff is unnecessary

Complications

Real-world complications

Several factors complicate carbon tariff design:

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Several factors complicate carbon tariff design:

1. **Resource shuffling:** clean production exported, dirty production consumed domestically
2. **Non-price policies:** regulations vs. carbon prices
3. **Export rebates:** should exports get carbon tax rebates?
4. **Retaliation:** trading partners may impose counter-tariffs

Resource shuffling

Suppose country j has both clean and dirty production

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- Export the clean stuff to country i
- Consume the dirty stuff domestically

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This is **another form of leakage**

Non-price policies

Many countries use regulations rather than carbon prices:

- Fuel economy standards
- Renewable portfolio standards
- Building codes

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How do you calculate the "equivalent carbon price"?

Very difficult: may require complex modeling

Export rebates

Should country i rebate carbon taxes on exports?

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- Symmetric with import tariffs

Export rebates

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Arguments for:

- Maintains competitiveness
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Arguments against:

- Reduces domestic incentive to decarbonize
- May look like an export subsidy (WTO issues)

WTO compatibility

Carbon tariffs face WTO scrutiny:

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EU CBAM designed carefully to fit within WTO rules

Key takeaways

Summary

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1. **Carbon tariffs** aim to address leakage and competitiveness concerns
2. **Carbon tax + tariff** is equivalent to a consumption tax
 - Reduces consumption of carbon-intensive goods
 - Can improve welfare if environmental benefit exceeds distortion costs
3. **Carbon tariff alone** = consumption tax + production subsidy
 - May increase domestic emissions
 - Less effective than tax + tariff

Summary (continued)

1. **World carbon tax** is first-best
 - Carbon tariffs are second-best when global coordination fails

Summary (continued)

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2. **Different carbon intensities** create measurement challenges

- Setting tariff = foreign intensity is efficient but looks protectionist
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Summary (continued)

1. **World carbon tax** is first-best

- Carbon tariffs are second-best when global coordination fails

2. **Different carbon intensities** create measurement challenges

- Setting tariff = foreign intensity is efficient but looks protectionist
- Setting tariff = domestic intensity under-prices foreign emissions

3. **Many complications** in practice

- Resource shuffling, non-price policies, export rebates, WTO rules

The bottom line

Carbon tariffs are a **complement** to carbon pricing, not a substitute

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- Set based on actual carbon content
- Part of broader international coordination

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Alone, they are an imperfect tool that may create new distortions